

Group 10: Laboratory 4

Hu's invariant moments

Keegan Rolls (202004149)
Nicholas Philpott (202018248)

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1. Download the grayscale test image (*image.png*), calculate *moments.m*, and *Moment invariants.m* files from Brightspace under Lab 04 and save them in your working directory. Read the files and understand the implementation of calculating the seven invariant moments.
2. Read the image, obtain the size of the image and pad the image by one-fourth the image size in all directions with zeros, and display the padded image. Use this as the basis for next steps (referred to as *Im1*).
3. Create and display the spatial transformation matrices *T1* for translation, *T2* for scaling to 0.5 (of original size), *T3* to rotate the image by 45 degrees, *T4* for rotating the image 90 degrees.
4. Transform *Im1* with *T1* and display the translated *Im2*
5. Transform *Im1* with *T2* and display the scaled *Im3*:
6. Transform *Im1* with *T3* and display the rotated *Im4*:
7. Transform *Im1* with *T4* and display the rotated *Im5*:
8. Flip the original image *Im1* from left to right using following command, generate and display flipped
9. The file *Moment_invariant.m* contains a function *Moment_invariant()*, which calls another function named *calculate_moments*. Collectively, these functions calculate the moment invariants. Use the function *Moment_invariant()* with images *Im1* to *Im6* (e.g., *Moment_invariants(Im1);*) to calculate the moment invariants for each image. Report the moments for each image in a table.
10. Discuss the values of the moment invariants computed for each image, including any similarities or differences. What do these similarities or differences mean in terms of image features?